
Apex Metropolitan Medical Centre: Master Operational Directory

The following directory outlines the core medical staff, their specialized departments, areas of clinical focus, and their standard weekly consultation schedules.

Medical Staff and Departmental Assignments

| Doctor Name | Department | Clinical Specialization | Room | Consultation Timings |

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| **Dr. John Doe** | Internal Medicine | Preventive Care, Metabolic Disorders | 101 | Mon-Fri: 09:00 - 13:00, 14:00 - 17:00 |

| **Dr. Sarah Jenkins** | Pediatrics | Neonatology, Early Childhood Development | 104 | Mon, Wed, Fri: 08:30 - 14:30 |

| **Dr. Rajesh Kumar** | Cardiology | Interventional Cardiology, Heart Failure | 202 | Tue-Sat: 10:00 - 16:00 |

| **Dr. Emily Chen** | Neurology | Movement Disorders, Stroke Rehabilitation | 205 | Mon-Thu: 09:00 - 15:00 |

| **Dr. Michael Thorne** | Orthopedics | Sports Medicine, Joint Replacement | 301 | Mon, Wed, Thu: 11:00 - 18:00 |

| **Dr. Anita Desai** | Dermatology | Clinical Dermatology, Dermato-Oncology | 304 | Tue-Fri: 09:30 - 13:30 |

| **Dr. Marcus Vance** | Psychiatry | Cognitive Behavioral Therapy, Addiction | 402 | Mon, Tue, Thu: 13:00 - 19:00 |

| **Dr. Elena Rostova** | Oncology | Medical Hematology, Solid Tumors | 405 | Mon-Fri: 08:00 - 14:00 |

Procedural and Diagnostic Departments

Beyond outpatient consultations, the hospital operates several high-capacity diagnostic wings. The Radiology department, overseen by Chief Radiologist Dr. William Vance, handles MRI, CT, and fluoroscopy imaging. The Pathology Laboratory operates 24/7, processing hematology, biochemistry, and microbiology panels to support the clinical staff.

Comprehensive Financial and Billing Structure

The financial framework at Apex Metropolitan is tiered based on the physician's specialization level and the complexity of the diagnostic procedures. All front-desk staff must adhere to these schedules when generating estimates and billing patients.

Outpatient Consultation Fees

Provider Tier	Qualifications / Role	Initial Consultation Fee	Follow-Up Fee (Within 14 Days)
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General Physician	MBBS, MD (Internal Medicine)	₹800	₹400
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Specialist	MD, MS, DNB	₹1,200	₹600
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Super-Specialist	DM, MCh, Fellowship	₹1,800	₹900
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Emergency Triage	ER Attending	₹1,500	N/A
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Dietician / Nutrition	BSc, MSc Clinical Nutrition	₹600	₹300
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Standard Diagnostic and Pathology Fees

Test / Investigation	Department	Standard Rate	Turnaround Time
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Complete Blood Count (CBC)	Pathology	₹400	4 Hours
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Lipid Panel	Pathology	₹750	6 Hours
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HbA1c (Glycosylated Hemoglobin)	Pathology	₹600	6 Hours
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Thyroid Profile (T3, T4, TSH)	Pathology	₹900	8 Hours
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X-Ray (Chest PA View)	Radiology	₹500	1 Hour
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Ultrasound (Whole Abdomen)	Radiology	₹1,500	2 Hours
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MRI Brain (Plain)	Radiology	₹7,500	24 Hours
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CT Scan (Abdomen & Pelvis)	Radiology	₹5,500	12 Hours
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ECG (Electrocardiogram)	Cardiology	₹350	Immediate
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2D Echocardiography	Cardiology	₹2,200	4 Hours
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Minor Outpatient Procedures

Procedure Description	Associated Department	Base Cost
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Simple Laceration Suture	Emergency / Surgery	₹1,500
Cast Application (Fiberglass)	Orthopedics	₹2,500
Joint Aspiration (Arthrocentesis)	Orthopedics / Rheumatology	₹3,000
Skin Biopsy (Punch/Shave)	Dermatology	₹2,800
Ear Wax Removal (Syringing)	ENT	₹800

Front Desk Standard Operating Procedure (SOP)

This manual is assigned to **Priya**, the Lead Outpatient Receptionist, and her team. It dictates the mandatory workflows for patient intake, data capture, and clinic flow management.

1. Patient Registration and Data Capture

All new patients must be entered into the Electronic Health Record (EHR) system. The mandatory fields are:

- * Full Legal Name
- * Date of Birth (DOB) in DD/MM/YYYY format
- * Primary Mobile Number (Verified via OTP)
- * Emergency Contact Information
- * Government ID Number (Aadhar/PAN)

> **Priya's Directive:** "Never bypass the mobile verification step. If a patient shares an older file without a verified number, update it immediately. Accurate demographics prevent duplicate EHR files and ensure lab results are routed via SMS automatically."

2. Triage and Walk-In Protocols

Patients arriving without an appointment must be evaluated at the desk.

Emergency indicators: Chest pain, severe breathlessness, profuse bleeding, or sudden paralysis require immediate transfer to the ER. Do not process out-patient billing; page the triage nurse immediately.

Routine walk-ins: Check the daily schedule. If a doctor has a cancellation, slot the walk-in. If fully booked, offer an appointment with the on-call General Physician (usually Dr. John Doe).

3. Billing and Insurance Workflow

1. Verify the patient's appointment in the scheduling module.

2. Confirm if the visit is a First-Time Consultation or a Follow-Up.
3. If the patient is utilizing corporate insurance, collect the TPA (Third Party Administrator) pre-authorization card.
4. Generate the invoice and collect co-pays before the patient proceeds to the vitals station.

Master Outpatient Appointment Registry: Tuesday Clinic

The following is Priya's master log for the morning shift. It contains the demographic data, assigned provider, and chief complaints for the day's scheduled patients.

| Patient Name | Date of Birth | Mobile Number | Assigned Doctor | Time | Chief Complaint |

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| **Kiran Rathore** | 18/08/1999 | 9874563215 | Dr. John Doe | 09:15 AM | Unexplained fatigue, joint pain |

| **Amit Sharma** | 12/04/1985 | 9988776655 | Dr. Rajesh Kumar | 10:00 AM | Post-op bypass follow-up |

| **Maria Garcia** | 22/11/1990 | 9123456780 | Dr. Anita Desai | 10:30 AM | Severe cystic acne flare-up |

| **James Holden** | 05/01/1972 | 9871234560 | Dr. Michael Thorne | 11:00 AM | Chronic right knee stiffness |

| **Aisha Khan** | 14/09/2015 | 9900112233 | Dr. Sarah Jenkins | 11:15 AM | Persistent nocturnal cough |

| **David Miller** | 30/10/1965 | 9012345678 | Dr. John Doe | 11:45 AM | Hypertension medication refill |

| **Divya Patel** | 02/02/1988 | 9112233445 | Dr. Emily Chen | 12:15 PM | Migraine with visual aura |

| **Samuel Jackson** | 17/07/1954 | 9223344556 | Dr. Elena Rostova | 12:30 PM | Chemotherapy port check |

| **Chloe Davis** | 25/12/2001 | 9334455667 | Dr. Marcus Vance | 13:00 PM | Generalized anxiety evaluation |

| **Wei Chen** | 08/08/1979 | 9445566778 | Dr. Rajesh Kumar | 14:00 PM | Palpitations during exercise |

| **Fatima Zahra** | 19/03/1995 | 9556677889 | Dr. Anita Desai | 14:30 PM | Suspected fungal dermatitis |

| ****Vikram Singh**** | 04/11/1982 | 9667788990 | Dr. Michael Thorne | 15:00 PM | Rotator cuff tear evaluation |

| ****Elena Rostova**** | 11/05/2010 | 9778899001 | Dr. Sarah Jenkins | 15:30 PM | Pre-camp physical exam |

| ****Robert Blake**** | 29/09/1960 | 9889900112 | Dr. John Doe | 16:00 PM | Type 2 Diabetes quarterly review |

| ****Nina Dobrev**** | 14/02/1993 | 9990011223 | Dr. Emily Chen | 16:30 PM | Carpal tunnel syndrome check |

Clinical Transcripts and EHR Encounter Logs

The following section contains detailed transcriptions and EHR data entries for selected patient encounters from the daily registry. This demonstrates the flow from the front desk with Priya to the clinical evaluation with the physician.

Encounter 1: Kiran Rathore (Internal Medicine)

****Time:**** 09:10 AM

****Location:**** Main Reception Desk

****Priya (Receptionist):**** "Good morning, welcome to Apex Medical. Do you have an appointment?"

****Kiran Rathore:**** "Yes, I have a 9:15 appointment with Dr. John Doe. My name is Kiran Rathore."

****Priya:**** "Thank you, Kiran. Let me pull up your file. I see you are a new patient. Could I please have your date of birth and mobile number for the registry?"

****Kiran Rathore:**** "Sure, my date of birth is the 18th of August, 1999. The number is 9874563215."

****Priya:**** "Perfect, I have that recorded: 18/08/1999, mobile 9874563215. Since this is your first visit, the initial consultation fee for General Medicine is ₹800. Will you be paying by card or UPI?"

****Kiran Rathore:**** "UPI is fine."

****Priya:**** "Thank you. Your payment is confirmed. Please take this token and wait in the blue seating area. The nurse will call you shortly to check your vitals before you see Dr. Doe in Room 101."

****Time:**** 09:25 AM

****Location:**** Room 101 - Internal Medicine

****Provider:**** Dr. John Doe

****Patient:**** Kiran Rathore

****Dr. John Doe:**** "Good morning, Kiran. I see from the nurse's intake that your blood pressure and temperature are perfectly normal, but you've been dealing with some persistent fatigue and joint pain. How long has this been going on?"

****Kiran Rathore:**** "Good morning, Dr. Doe. It started about a month ago. I wake up feeling exhausted, even if I sleep for eight or nine hours. And over the last two weeks, my knees and wrists have felt stiff and achy, especially in the mornings."

****Dr. John Doe:**** "I understand. Fatigue combined with joint stiffness in a 26-year-old can point to a few different things. Have you had any recent fevers, unexplained weight loss, or rashes?"

****Kiran Rathore:**** "No fevers. Maybe a little bit of weight loss, but nothing major. I haven't noticed any rashes."

****Dr. John Doe:**** "How about your diet and lifestyle? Have there been any major changes, increased stress, or lack of sun exposure?"

****Kiran Rathore:**** "I work in IT, so I'm indoors most of the day in front of a screen. Work has been a bit stressful lately."

****Dr. John Doe:**** "Let me do a quick physical examination. I'll check your joints for swelling and test your range of motion."

(Dr. Doe performs a physical examination, palpating the wrists and knees. He checks for effusions and restricted movement.)

****Dr. John Doe:**** "There's no obvious swelling or heat in the joints, which is good. The stiffness might be mechanical, or it could be related to a systemic deficiency. Given your indoor lifestyle, Vitamin D deficiency is a strong suspect—it commonly causes deep bone ache and severe fatigue. We also need to rule out early markers of autoimmune issues or basic anemia."

****Kiran Rathore:**** "So what's the next step?"

****Dr. John Doe:**** "I'm going to order a comprehensive blood panel. We will check your Complete Blood Count, Vitamin D3 levels, B12, Thyroid function, and a C-Reactive Protein test to check for general inflammation. Please take this requisition form back to Priya at the front desk, and she will direct you to the Pathology lab."

****Kiran Rathore:**** "Alright, thank you, Doctor. When should I follow up?"

****Dr. John Doe:**** "The results should be uploaded to your patient portal by tomorrow evening. Book a follow-up appointment with Priya for Thursday. We'll go over the numbers and get you on a treatment plan."

EHR SOAP Note: Kiran Rathore

*** **Subjective:**** 26-year-old male presents with a 4-week history of generalized fatigue, unrefreshing sleep, and bilateral morning stiffness in knees and wrists. Denies fever, rash, or significant weight loss. Sedentary indoor occupation.

*** **Objective:**** Vitals stable (BP 118/76, HR 72, Temp 98.6F). Joint exam reveals mild tenderness over radiocarpal joints bilaterally without active synovitis, warmth, or effusion. Knee ROM intact, negative for crepitus.

*** **Assessment:**** Generalized fatigue and arthralgia. High suspicion for Vitamin D deficiency vs. viral post-infectious fatigue vs. early inflammatory arthropathy.

*** **Plan:**** Ordered CBC, Vit D3 (25-OH), Vit B12, TSH, and CRP. Advised adequate hydration and gentle stretching. Patient to follow up in 48 hours with lab results.

Encounter 2: Aisha Khan (Pediatrics)

****Time:**** 11:05 AM

****Location:**** Main Reception Desk

****Priya (Receptionist):**** "Hello Mrs. Khan. I see Aisha is here for her 11:15 with Dr. Jenkins."

****Mrs. Khan (Mother):**** "Yes, she has been coughing all night. I booked this morning."

****Priya:**** "I have her file right here. Aisha Khan, date of birth 14/09/2015. Since she's an existing patient and hasn't been seen in the last 14 days, the standard pediatric consultation fee of ₹1,200 applies. You can proceed directly to Room 104 after payment; the nurse is ready for her weight check."

****Time:**** 11:20 AM

****Location:**** Room 104 - Pediatrics

****Provider:**** Dr. Sarah Jenkins

****Patient:**** Aisha Khan (Accompanied by mother)

****Dr. Sarah Jenkins:**** "Hello Aisha! Hi Mrs. Khan. I hear we've got a pesky cough keeping everyone awake. Tell me more about it."

****Mrs. Khan:**** "It started three days ago as a runny nose, but yesterday it moved into her chest. It gets much worse when she lies down at night. It sounds very dry and hacking."

****Dr. Sarah Jenkins:**** "Has she been running a fever or complaining of a sore throat?"

****Mrs. Khan:**** "A mild fever yesterday, about 100.2, but it broke with paracetamol. She says her throat hurts from the coughing."

****Dr. Sarah Jenkins:**** "Alright, Aisha, let's take a listen to your lungs. Take a deep breath for me like you're blowing out birthday candles."

(Dr. Jenkins auscultates the patient's lungs anteriorly and posteriorly.)

****Dr. Sarah Jenkins:**** "Her lungs are quite clear. I don't hear any wheezing or crackles, which means the infection hasn't settled deep into the chest. Her throat is a bit red, and I can see some post-nasal drip. That's what's causing the nighttime cough—the mucus drips down the back of her throat when she lies flat, irritating her airway."

****Mrs. Khan:**** "That's a relief. Do we need antibiotics?"

****Dr. Sarah Jenkins:**** "No, this looks entirely viral. Antibiotics won't help and might just upset her stomach. I recommend elevating her head slightly at night with an extra pillow. I'll prescribe a gentle antihistamine syrup to dry up the nasal drip before bedtime, and a saline nasal spray. Keep her hydrated with warm fluids."

EHR SOAP Note: Aisha Khan

* ****Subjective:**** 10-year-old female presents with 3 days of rhinorrhea, progressing to dry, hacking nocturnal cough. Low-grade fever (100.2F) resolved with antipyretics. No shortness of breath.

* ****Objective:**** Chest is clear to auscultation bilaterally. No wheezes, rales, or rhonchi. Pharynx is erythematous with visible clear post-nasal drip. Tympanic membranes are clear.

* ****Assessment:**** Viral upper respiratory infection (URI) with post-nasal drip causing reactive cough.

* ****Plan:**** Prescribed Cetirizine syrup 5mg at bedtime for 5 days. Recommended saline nasal irrigation and supportive care. Return if fever spikes >102F or if respiratory distress develops.

Encounter 3: James Holden (Orthopedics)

****Time:**** 10:55 AM

****Location:**** Main Reception Desk

****Priya (Receptionist):**** "Good morning, Mr. Holden. Checking in for Dr. Thorne?"

****James Holden:**** "Yes, Priya. My right knee is acting up again."

****Priya:**** "I'm sorry to hear that. Your date of birth is 05/01/1972, mobile 9871234560, correct?"

****James Holden:**** "That's right."

****Priya:**** "This falls under a follow-up visit since you saw Dr. Thorne last week for the initial evaluation. The follow-up fee is ₹600. Dr. Thorne is waiting for you in Room 301."

****Time:**** 11:10 AM

****Location:**** Room 301 - Orthopedics

****Provider:**** Dr. Michael Thorne

****Patient:**** James Holden

****Dr. Michael Thorne:**** "James, good to see you. I have the MRI results for your right knee right here on the monitor. Let's take a look."

****James Holden:**** "Give it to me straight, Doc. Am I looking at surgery?"

****Dr. Michael Thorne:**** "Not immediately. Let me walk you through the imaging. As we suspected, there is moderate osteoarthritis in the medial compartment of the knee—the cartilage is thinning, which is causing that bone-on-bone friction and stiffness. However, your meniscus and ligaments are completely intact."

****James Holden:**** "Why does it hurt so much when I try to walk down stairs?"

****Dr. Michael Thorne:**** "Because of the inflammation. The joint capsule gets irritated from the friction. We have a few conservative options before we even discuss joint replacement. I'd like to start with an intra-articular corticosteroid injection. It will deliver powerful anti-inflammatory medication directly into the joint space."

****James Holden:**** "Will that cure it?"

****Dr. Michael Thorne:**** "It's not a cure for the worn cartilage, but it can provide significant pain relief for several months. During that time, I want you in physical therapy. Strengthening the quadriceps and hamstrings will take the mechanical load off the knee joint itself."

****James Holden:**** "Let's do the injection. I need to get back to work without limping."

****Dr. Michael Thorne:**** "We can do that today. I'll have the nurse prep the tray. It's a quick outpatient procedure. Priya will add the procedure fee to your bill on your way out."

EHR SOAP Note: James Holden

****Subjective:**** 54-year-old male returns to discuss MRI results for chronic right knee pain. Pain is localized to medial joint line, exacerbated by descending stairs.

****Objective:**** MRI Right Knee (reviewed): Grade II-III chondromalacia of the medial femoral condyle and medial tibial plateau. Menisci and cruciate ligaments are intact. Small joint effusion noted.

****Assessment:**** Medial compartment osteoarthritis, right knee.

****Plan:**** Discussed conservative management versus surgical intervention. Patient elected for corticosteroid injection. Consented and performed right knee intra-articular injection (40mg Triamcinolone + 1% Lidocaine) using aseptic technique. Tolerated procedure well. Prescribed 6 weeks of targeted physical therapy. Follow up in 2 months.

Administrative End-of-Day Procedures

At the close of the clinic day, the front desk operations switch from patient-facing tasks to financial reconciliation and data auditing.

Receptionist Checklist (Priya)

- **Reconciliation:**** Tally the physical cash, card receipts, and UPI transfers against the EHR billing module. The total collected must match the sum of Consultation Fees + Procedure Fees + Diagnostic Fees generated for the day.
- **No-Show Follow-up:**** Any patient who failed to arrive for their scheduled slot (e.g., if Vikram Singh missed his 15:00 appointment) must be contacted via their registered mobile number to reschedule.
- **Lab Report Routing:**** Ensure that all automated SMS alerts were successfully dispatched to patients whose lab results were finalized by the Pathology department. (e.g., verifying that Kiran Rathore's mobile, 9874563215, received the secure link to his blood panel results).
- **Handoff:**** Prepare the appointment matrix for the following day, ensuring all doctors have a printed manifest of their scheduled patients. Ensure the ER desk is briefed on any out-patient cases that might return with complications.

1. Reference Doctor Schedule (Context)

Doctor Name	Department	Specialty	Room	Schedule
Dr. John Doe	Internal Medicine	Preventive Care, Metabolic Disorders	101	Mon-Fri: 09:00-13:00, 14:00-17:00
Dr. Sarah Jenkins	Pediatrics	Neonatology, Early Childhood Development	104	Mon, Wed, Fri: 08:30-14:30
Dr. Rajesh Kumar	Cardiology	Interventional Cardiology, Heart Failure	202	Tue-Sat: 10:00-16:00
Dr. Emily Chen	Neurology	Movement Disorders, Stroke Rehabilitation	205	Mon-Thu: 09:00-15:00
Dr. Michael Thorne	Orthopedics	Sports Medicine, Joint Replacement	301	Mon, Wed, Thu: 11:00-18:00
Dr. Anita Desai	Dermatology	Clinical Dermatology, Dermato-Oncology	304	Tue-Fri: 09:30-13:30
Dr. Marcus Vance	Psychiatry	Cognitive Behavioral Therapy, Addiction	402	Mon, Tue, Thu: 13:00-19:00
Dr. Elena Rostova	Oncology	Medical Hematology, Solid Tumors	405	Mon-Fri: 08:00-14:00

2. Conversational Queries (Detailed Input)

Scenario: Requesting an appointment before operating hours

User: "I need to book an appointment with Dr. John Doe at 8:00 AM next Monday."

Assistant: "Dr. John Doe is not available at 8:00 AM. His Monday hours are 09:00 - 13:00 and 14:00 - 17:00 in Room 101. Would you like to schedule for 09:00 AM instead?"

Status: Out of Hours Rejection / Doctor: Dr. John Doe

Scenario: Successful booking by specialty

User: "Can I see a cardiologist this Wednesday around 11:30 AM?"

Assistant: "Yes, Dr. Rajesh Kumar (Cardiology) is available on Wednesdays. I have confirmed your appointment for Wednesday at 11:30 AM in Room 202."

Status: Valid Booking / Doctor: Dr. Rajesh Kumar

Scenario: Requesting an appointment on a day the doctor is off

User: "Schedule an appointment for my child with Dr. Sarah Jenkins on Tuesday morning."

Assistant: "Dr. Sarah Jenkins (Pediatrics) is not in the clinic on Tuesdays. Her available days are Monday, Wednesday, and Friday from 08:30 to 14:30 in Room 104. Which of those days works best for you?"

Status: Day Off Rejection / Doctor: Dr. Sarah Jenkins

Scenario: Successful booking by doctor name

User: "Book me with Dr. Michael Thorne on Thursday at 2:00 PM for a sports injury."

Assistant: "Your appointment is confirmed. You are scheduled with Dr. Michael Thorne (Orthopedics) on Thursday at 14:00 in Room 301."

Status: Valid Booking / Doctor: Dr. Michael Thorne

Scenario: Requesting an appointment during a lunch break/gap

User: "I'd like to see Dr. John Doe on Friday at 1:30 PM for a checkup."

Assistant: "Dr. John Doe is unavailable at 13:30 on Fridays as he is out of the office between 13:00 and 14:00. I can book you at 14:00. Would that work?"

Status: Lunch Gap Rejection / Doctor: Dr. John Doe

3. Direct/Terse Queries (Minimal Input)

Scenario: Perfect Match (Valid Booking)

User: "Dr. Emily Chen, Monday, 10:00 AM."

Assistant: "Appointment confirmed. You are scheduled with Dr. Emily Chen (Neurology) on Monday at 10:00 AM in Room 205."

Status: Valid Booking / Doctor: Dr. Emily Chen

Scenario: Time is Outside Working Hours

User: "Dr. Anita Desai, Wednesday, 2:00 PM."

Assistant: "Dr. Anita Desai is not available at 2:00 PM. Her Wednesday hours are from 09:30 to 13:30 in Room 304. Would you like an appointment within that timeframe?"

Status: Out of Hours Rejection / Doctor: Dr. Anita Desai

Scenario: Doctor is Not in the Clinic

User: "Dr. Marcus Vance, Friday, 4:00 PM."

Assistant: "Dr. Marcus Vance (Psychiatry) does not consult on Fridays. His available days are Monday, Tuesday, and Thursday from 13:00 to 19:00. Shall I book you for one of those days?"

Status: Day Off Rejection / Doctor: Dr. Marcus Vance

Scenario: Booking in the Afternoon Block

User: "Dr. John Doe, Tuesday, 2:30 PM."

Assistant: "Appointment confirmed. You are scheduled with Dr. John Doe (Internal Medicine) on Tuesday at 2:30 PM in Room 101."

Status: Valid Booking / Doctor: Dr. John Doe